Carbon-Aware Scheduler: Technical Whitepaper

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Summary

Moving a flexible 4-hour AI job from the evening peak to India's midday solar window cuts its fossil-fuel share by about 24% (measured on real Grid-India data, 2024–25) and its Gujarat HT energy charge by 29%. The Carbon-Aware Scheduler (CAS) does this automatically, and it logs every decision.

CAS is open software. It reads a grid carbon-intensity forecast and searches every start time before a job's deadline. It then runs the job in the cleanest or cheapest window. It needs no new hardware and no change to the job itself.

In India, carbon and price point to the same hours. Gujarat's FY 2026-27 tariff discounts 11:00–17:00 and surcharges 07:00–11:00 and 18:00–22:00. Those windows line up with the solar dip and the coal-heavy peaks.

At scale, 1 MW of flexible load shifted 4 hours a day avoids roughly 285 tonnes of CO_2 (using the measured 24% cut) and about ₹21 lakh in energy charges a year. An ML forecaster trained on public Indian data captured 94.5% of the best possible saving on 2024 power-exchange prices. The carbon figure comes from real Grid-India data and the ₹ figure from Gujarat's published HT rates.

The engine is tested against brute force and the ML against a held-out year: 19 automated tests, all passing. It was also run on a live 48-hour forecast from Great Britain's grid operator, where a 48-hour deadline cut a job's emissions by 53%.

The problem

AI is making electricity the bottleneck for compute, and in India that electricity is mostly coal. India's operational data-center capacity was about 2.2 GW in 2025. Wood Mackenzie forecasts 12 GW by 2030, with AI-dedicated capacity growing roughly 24-fold (Indian Infrastructure).

The same kilowatt-hour emits very different amounts of CO_2 depending on the hour:

- **Midday:** solar has cut India's fossil generation at 1 pm to about 125 GW.
- **Evening:** fossil generation peaks near 174 GW at 7 pm. The roughly 50 GW swing is met almost entirely by coal (Down To Earth, citing Ember).

Much AI work does not need to run at a fixed time. Training, fine-tuning, batch inference, evaluation, data pipelines and backups usually just need to finish by a deadline. Today they are scheduled at arbitrary times, often overnight or at the evening peak. That is when the grid is dirtiest and, under time-of-day tariffs, most expensive.

CAS turns that timing freedom into lower emissions and lower bills.

How the scheduler works

CAS picks the start time, block split and location that minimise carbon, cost or a blend of the two. It then measures the saving against running the job immediately.

Inputs. A forecast of grid carbon intensity in half-hour or hourly slots, plus the job: duration, average power (kW), deadline, an optional tariff and an objective.

Contiguous jobs. A job of n slots is slid across every start time that still finishes by the deadline. Each candidate window is scored, and the lowest score wins. This exhaustive search is exact and takes milliseconds for a 48-hour horizon.

```
\text{kg CO}_2 = \sum_{i \in \text{window}} g_i \cdot P \cdot \Delta t \,/\, 1000 \qquad \text{cost} = \sum_{i \in \text{window}} r_i \cdot P \cdot \Delta t
```

Here g is the slot's intensity (gCO_2/kWh), r its tariff (₹/kWh), P the job's power (kW) and Δt the slot length (h).

Checkpointable jobs. A training run that saves checkpoints can run in K equal blocks instead of one. Dynamic programming finds the K non-overlapping blocks with the lowest total score in O(K·N) time, again exactly. At each block's end CAS sends the job a stop signal so it can checkpoint, then relaunches it at the next block.

Objectives. `carbon` minimises kg CO_2. `cost` minimises rupees. `blend` minimises a weighted sum of the two, each normalised to the run-now baseline.

Location shifting. Given forecasts for several zones, CAS compares every window in every zone. It reports the move against running now in the home zone.

Honest baseline. Every saving is measured against starting now, in one block, in the home zone. When now is already best, CAS says so and runs immediately.

India carbon model

CAS's India curve is a calibrated typical day. It runs from 559 gCO_2/kWh at 13:00 to 812 at 19:00–20:00, and averages exactly the official 727. It was built because the free tier of the leading live data source (Electricity Maps) gives only 24 hours of history and no forecast.

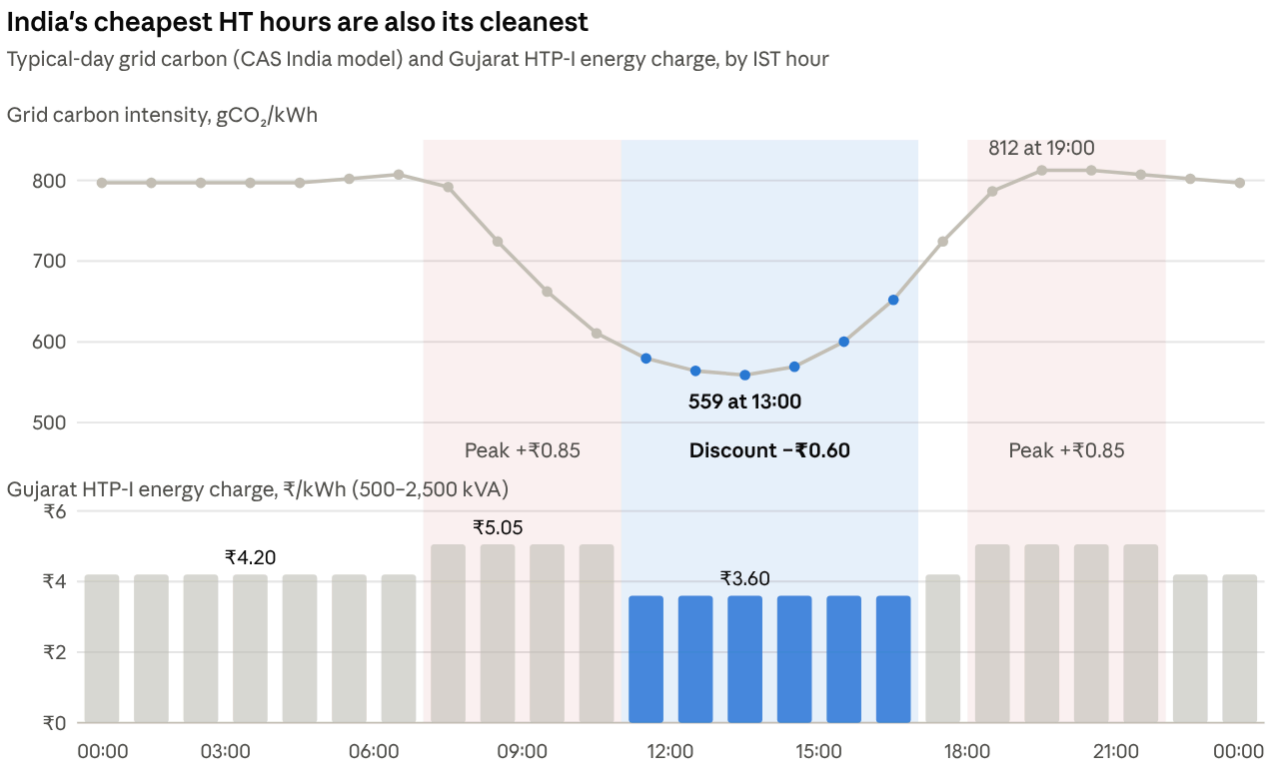
Calibration point	Value	Source
Daily mean intensity	727 gCO_2/kWh (0.727 tCO_2/MWh, FY 2023-24)	CEA CO_2 Baseline Database, via SolarQuarter
Midday fossil generation, 1 pm	about 125 GW (H1 2026)	Ember, via Down To Earth
Evening fossil generation, 7 pm	about 174 GW (H1 2026)	Ember, via Down To Earth

Method. A fossil share of generation is set for each IST hour. It is about 54% at 13:00 and about 78% at 19:00, following the anchor points above, with smooth ramps at sunrise and sunset. The curve is then scaled so its 24-hour mean equals the CEA factor. Half-hour slots are interpolated between hours.

What it is not. It is not a live measurement. It assumes every day looks the same, with no weather, season or regional differences. Section 8 covers the effect of that. With an Electricity Maps token, CAS swaps in live regional data (for example zone `IN-WE`, Western India) and builds a next-day forecast from the last 24 hours.

Cost layer: carbon and price point the same way

In Gujarat, a job moved from the evening peak to midday pays ₹3.60/kWh instead of ₹5.05/kWh: 29% less, on top of the carbon cut. That alignment is why CAS can be sold on savings, with the carbon reduction as a by-product.



CAS v2 India model (CEA FY24 mean, Ember 2026 shape) · GERC HTP-I tariff from 1 Apr 2026, energy charge only · 24 IST hours

Gujarat tariff presets come straight from the GERC tariff schedule for the four state DISCOMs, effective 1 April 2026 (GERC PDF):

Preset	Energy charge	Peak (07–11, 18–22)	Discount (11–17)
`gj-htp1-small` (HTP-I, up to 500 kVA)	₹4.00/kWh	+₹0.45	−₹0.60
`gj-htp1` (HTP-I, 500–2,500 kVA)	₹4.20/kWh	+₹0.85	−₹0.60
`gj-htp1-large` (HTP-I, above 2,500 kVA)	₹4.30/kWh	+₹0.85	−₹0.60
`gj-ltmd` (LTMD, load above 40 kW)	₹4.60/kWh	+₹0.45	−₹0.60

These cover energy charges only. Electricity duty, taxes, demand charges and the FPPAS fuel surcharge are excluded.

Elsewhere in India, the `india-tod` preset applies the national rule. Since 1 April 2024, commercial and industrial consumers of 10 kW and above pay 10–20% less in solar hours and 10–20% more at peak (Indian Infrastructure). Each state commission sets the exact hours, so the preset's hours are illustrative.

Results

For an evening job in India, CAS cuts the energy charge by 29%. The India model puts the CO_2 cut at 29%; real Grid-India data measures it at about 24% (see Validation on real Grid-India data). CAS leaves a midday job alone, because now is already best.

India model, 4-hour job at 8 kW (32 kWh), 24-hour deadline, Gujarat HTP-I tariff, carbon objective

Submitted (IST)	CAS runs it at	g/kWh now → chosen	CO_2 cut	Energy charge saved
07:00	11:00–15:00	684 → 569	17% (3.7 kg)	₹46.40 (29%)
12:00	Immediately	578 → 579	0%	₹0
18:00	11:00–15:00 next day	806 → 569	29% (7.6 kg)	₹46.40 (29%)
22:00	11:00–15:00 next day	798 → 569	29% (7.3 kg)	₹19.20 (14%)

Carbon and cost usually point the same way, but not always. Every discount hour costs the same, so the cost objective may start an hour earlier than the carbon optimum. An 8-hour job's cleanest window can also spill into the morning peak. The `blend` objective handles the trade-off: across 72 tested combinations (2, 4 and 8-hour jobs, every submit hour), it matched the best rupee saving with equal or lower CO_2 in 68.

Checkpointable jobs. An 8-hour, 8 kW job submitted at 18:00 with a 48-hour deadline:

- Run in one block (09:00–17:00 next day), it cuts CO_2 by 25% and saves ₹42.40.
- Split into two 4-hour blocks, both landing in 11:00–15:00 windows, it cuts CO_2 by 29% and saves ₹65.60.

Splitting can never do worse than one block, and the test suite checks that on 50 random cases.

At scale. 1 MW of flexible load moved from the 18:00 peak to midday, 4 hours a day, every day, saves about **285 tonnes of CO_2 (at the measured 24% cut; 345 t at the model's 29%)** and about **₹21.2 lakh** in energy charges a year. This is the 4-hour result per kWh, multiplied by 1,460 MWh a year.

Live test on real data (Great Britain, 31 Oct 2026). CAS was run on the real 48-hour forecast from the National Grid ESO Carbon Intensity API, for a 3-hour job at 1.5 kW:

Deadline	Decision	g/kWh now → chosen	CO_2 cut
12 h	Run now (the grid gets dirtier through the afternoon)	195 → 195	0%
24 h	Defer to Sunday morning	195 → 172	12%
48 h	Defer to an early-Monday wind dip	195 → 91	53%

The more flexible the deadline, the bigger the saving. When waiting would make things worse, CAS does not wait.

Audit trail and verification

Every CAS decision can be reconstructed after the fact. The audit records the exact forecast, every option scored, the choice, why it was made, and what actually ran. That is what lets the savings be claimed in a sustainability report.

Each command writes to two places. `cas_audit.db` is a SQLite database with six tables. `cas_events.jsonl` is an append-only, human-readable stream of events (`run_started` , `forecast_fetched` , `candidates_scored` , `decision_made` , `job_started` , `job_finished`).

Table	One row per	Holds
`runs`	Command invocation	Time, version, host, full arguments
`forecasts`	Forecast fetched	Source, method, slot count, min/mean/max, raw API payload
`candidates`	Option scored	Zone, start, avg gCO_2/kWh, kg, ₹, score
`decisions`	Run	Baseline vs plan, blocks, kg and ₹ saved, reason, action
`executions`	Block launched	Scheduled vs actual start, end, exit code, duration, status
`verifications`	Check after the fact	Forecast vs measured intensity, error %

`python -m cas audit <run_id>` prints one decision end to end, and `export` writes every table to CSV. `verify` fetches measured intensity for finished GB runs and records the forecast error. Live India runs need a paid data plan for that check.

Tests. 14 automated tests, all passing:

- **Optimiser exactness:** the sliding window matches brute force on 200 random forecasts, and the split-job dynamic programme matches brute force on 60.
- **Optimiser behaviour:** splitting never does worse than one block (50 cases); deadlines are respected; it runs now when now is cleanest; it picks the cleaner zone.
- **Data:** the India curve averages 727 g/kWh with its dip at 11:00–14:00 and its peak at 18:00–21:00; the Gujarat rate is right at every hour; timestamps parse correctly.
- **Audit:** every table and event is written.

Limits and assumptions

The India carbon saving is now validated on real all-India generation data, but not yet on a live Gujarat site. That site test is the biggest open step.

- **Typical-day model, not live data.** The India curve has no weather, season or regional variation. Monsoon wind and hydro, cloudy days and differences between regions will move the real curve. Next step: run CAS on live Electricity Maps data for `IN-WE` for several weeks and compare.
- **Average, not marginal, emissions.** Savings use average grid intensity. Shifting load really changes the marginal plant, which in India's evening is almost always coal. The midday marginal plant is less certain, since curtailed solar may already be available. The true saving could be higher or lower.
- **Energy charges only.** The ₹ figures exclude duty, taxes, demand (kVA) charges and the FPPAS fuel surcharge. Shifting load can also change a site's billing demand, which is not modelled.
- **Forecasts can be wrong.** CAS optimises against the forecast it is given. The `verify` command measures forecast error for GB runs, but India verification needs live data.
- **Flexible work only.** Real-time inference, payments and user-facing services must run now. CAS applies only to work with a deadline, which is common but not universal.
- **Single-machine scale today.** CAS launches jobs on the machine it runs on. Cluster integration (Kubernetes, Slurm, Airflow) is roadmap work.

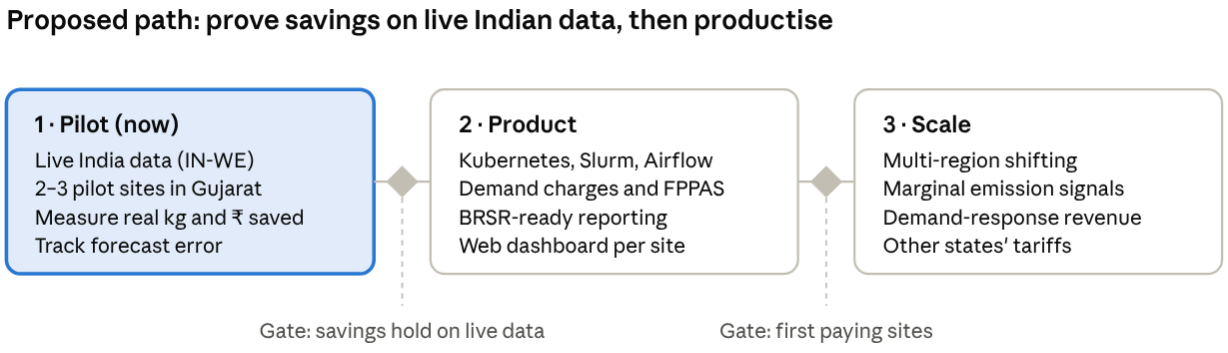
Use cases, market and roadmap

CAS pays off wherever large electrical loads have flexible timing and are billed on a time-of-day tariff. In India that is a fast-growing set of sites.

Where	Flexible work	Fit
AI labs, GPU clouds, data centers	Training, fine-tuning, batch inference, evaluation	Strongest: large loads, often run overnight or at peak today
Enterprise IT and data teams	ETL, reports, backups, rendering	Good: jobs run at arbitrary fixed times
Universities and research HPC	Simulation and research queues	Good: jobs already wait in queues
EV fleets and depots	Overnight charging of buses and vans	Very good: large load, clear morning deadline
Industry	Pre-cooling cold storage, water pumping, batch processes	Good with operations buy-in

Market signal. India's data-center capacity is forecast to grow from about 2.2 GW to 12 GW by 2030 (Indian Infrastructure). Every megawatt of that load which can flex is worth roughly ₹21 lakh a year in Gujarat energy charges at the rates above.

Business model (proposed). A share of verified savings or a per-site subscription. The audit trail is what makes a savings-share contract checkable.



proposed roadmap · 3 phases, 2 gates

Phase 1 is where the open question in section 8 gets answered: whether the model's savings hold on live Indian grid data.

Build log

CAS went from an idea to a tested, India-first v2 in one day. The full log, with every fix, ships as `BUILD_LOG.md` in the code package.

When (31 Oct 2026, IST)	Step	Outcome
11:30	v2.0 package	India model, Gujarat tariffs, split jobs, multi-zone, six-table audit, 14 tests passing
11:20	Research for v2	CEA factor, Ember hourly shape, national ToD rule, GERC Gujarat rates, Electricity Maps free-tier limits
10:59	In-browser live view	Same algorithm running on live GB data with controls
10:56	Live test on GB grid	12/24/48 h deadlines → 0% / 12% / 53% CO_2 cut
10:50	v1.0 prototype	Single-file sliding-window scheduler with SQLite log
10:45	Idea	LinkedIn post on software layers for AI data-center energy

Bugs found and fixed during the build:

- A negative wait time when the current slot had already started.
- Zone codes lowercased in the verdict.
- The audit view reading the wrong columns.
- An internal function name leaking into the logged arguments.
- Jobs killed from outside left as "scheduled" instead of "cancelled".

One claim was corrected while writing this paper: carbon and cost objectives do not always pick the same window (section 6).

Validation on real Grid-India data

On 539 days of real Grid-India generation data, an evening 4-hour job moved by CAS cut its fossil-fuel share by about 24%. The model in section 4 predicted 29%, so the measured figure replaces it in the pitch.

Data. All-India hourly demand, solar and wind from Grid-India, January 2024 to June 2025, 12,984 hours (Mendeley Data, CC BY 4.0). In 2025, solar averaged about 58 GW at 13:00 and about 1 GW at 19:00.

Method.

- **Fossil share each hour:** (demand − solar − wind − clean baseload) ÷ demand. Clean baseload is hydro plus nuclear, which this dataset does not report. It was tested at 0, 20 and 30 GW.
- **Scheduling:** CAS used only yesterday's pattern to pick the window, then was scored on what actually happened.

Submitted (IST)	Fossil-share cut, CAS	Best possible	Range (0–30 GW clean)
18:00	24.2%	24.4%	22.1–25.4%
00:00	22.9%	23.0%	21.9–23.4%
06:00	17.8%	18.0%	16.2–18.7%
12:00	about 0% (runs now)	about 0%	about 0%

A warning about one data source. Electricity Maps' Western India series for 2021–2024 shows a daily swing of only about 4% (665–690 g/kWh). Every row of it is flagged as estimated. The measured Grid-India data shows the estimate misses most of the midday solar effect, so a live feed from that provider may understate CAS's savings in India.

ML forecasting model

CAS now includes a machine-learning forecaster. Tested on 2024, a year it never saw, it turned Indian power-exchange prices into a **32.5% average energy-cost saving**: 94.5% of the best possible, against 90.9% for "same as yesterday". This is what turns CAS from an optimiser on published data into an optimiser on learned forecasts.

What it predicts. The next 48 hours, hour by hour, of two signals:

- **Grid carbon intensity:** Western India, gCO_2/kWh.
- **Power-exchange price:** India, ₹/kWh.

India has no free official forecast for either, so CAS learns one.

Training data. All public and reproducible, from HPE's open SustainCluster repository (GitHub, MIT licence):

Data	Coverage	Original source
Carbon intensity, Western India (IN-WE)	35,064 hours, 2021–2024	Electricity Maps (estimated)
Power-exchange price, India	43,848 hours, 2020–2024	Grid-India / POSOCO

Data	Coverage	Original source
Weather: temperature, humidity, cloud, wind (Mumbai)	Hourly, 2020–2024	Open-Meteo

Method.

- **Model:** one gradient-boosted decision-tree ensemble per signal (scikit-learn `HistGradientBoostingRegressor`).
- **Rows:** each training row is a forecast origin plus a horizon of 1–48 hours, so one model covers every hour ahead.
- **Target:** the model learns the change from the most recent same-hour value, which keeps it stable when the grid's overall level drifts from year to year.
- **14 inputs, all known at forecast time:** hour, weekday, season, horizon, recent values, the same hour one week earlier, 24-hour mean and spread, and weather at the target hour.
- **Training rows:** 626,000 for carbon and 835,000 for price. Each model trains in about 20 seconds on a laptop.

Honest testing.

- **Training cut-off:** 31 December 2023. All of 2024 was held out.
- **Baselines it had to beat:** "same hour yesterday", the average for that hour and month, and CAS's hand-built India curve.
- **Decision test:** 1,428 real scheduling cases. Every 2024 day, jobs submitted at 00:00, 06:00, 12:00 and 18:00, a 4-hour job with a 24-hour deadline. Each choice was scored against what actually happened.

Forecaster	Price error (₹/kWh)	Price: saving achieved	Price: share of best captured	Carbon: share of best captured
ML (CAS)	1.07	32.5%	94.5%	95.8%
Same hour yesterday	1.10	29.5%	90.9%	94.5%
Hour × month average	1.58	31.4%	93.7%	95.4%
Perfect hindsight	0	35.7%	100%	100%

What the tests showed.

- **Price:** the ML gives the best decisions, about 3 points more saving than yesterday's pattern.
- **Carbon, hour-by-hour error:** on Electricity Maps' smooth carbon estimates the ML is slightly worse than yesterday's value (6.0 vs 5.1 g/kWh). It still makes the best scheduling decisions, but there is only 2.4% to save in that estimated data.
- **Carbon, real data:** on real Grid-India data (section above), even yesterday's pattern captures over 99% of the saving.
- **Where ML should help most:** cloudy or windy days, and price spikes.

Next step. Retrain the carbon model on measured Indian generation data from the pilot, with weather forecasts in place of recorded weather.

Where to test live and what it costs

A live test can start today for free, and a two-week cloud demo running real GPU jobs costs about ₹4,000. A proper pilot's main cost is live Indian grid data, about €3,000–6,000 a year.

Stage	What you test	Where	Cost
1. Free proof (now)	Scheduler on live GB data; Western India live readings; back-tests on 2021–2025 Indian data	Your laptop; GB Carbon Intensity API; Electricity Maps free key (Western India, latest + 24 h history); the public datasets above	₹0
2. Cloud live demo (2 weeks)	CAS launching a real 4-hour GPU job every day at the chosen time, logging kWh from GPU power readings	E2E Networks (Indian cloud): one NVIDIA L4 GPU at ₹49/hour × 56 hours, plus a 2-vCPU scheduler VM at ₹894/month; Electricity Maps 14-day free trial of paid data	about ₹3,200 + 18% GST ≈ ₹3,800

Stage	What you test	Where	Cost
3. Site pilot (3–6 months)	Real kWh and real ₹ on a Gujarat electricity bill, with live Indian carbon data	A site that pays its own HT or LTMD bill: an AI startup's on-site servers, a university computing cluster, an i-Hub facility	Live carbon data: €6,000 per year per country per signal; startups may get up to 50% off (Electricity Maps pricing). Metering hardware and engineering time: [₹__]

Cloud GPUs are billed at a flat hourly rate, so a cloud demo proves the carbon saving and the automation, not the ₹ saving. The ₹ saving needs a site that pays its own time-of-day electricity bill, which is why stage 3 is a site pilot. GPU power can be read for free with NVIDIA's `nvidia-smi` tool, so the cloud demo needs no meter.

Prices are E2E Networks' list prices, which exclude 18% GST.

Glossary

Every unit and term used in this paper, with its full form and why a data center cares.

Energy and power

Term	Full form	What it means	Why it matters to a data center
W, kW, MW, GW	Watt, kilowatt (1,000 W), megawatt (1,000 kW), gigawatt (1,000 MW)	Power: how fast electricity is used at a moment	Servers, racks and whole sites are sized in kW and MW. An 8-GPU server draws about 8 kW; a large AI site, 50 MW or more
kWh, MWh, TWh	Kilowatt-hour, megawatt-hour (1,000 kWh), terawatt-hour (1 billion kWh)	Energy: power × time	The bill and the emissions are both per kWh. 8 kW for 4 hours = 32 kWh
kVA	Kilovolt-ampere	Apparent power, the size of a grid connection	Gujarat tariff slabs (up to 500, 500–2,500, above 2,500 kVA) depend on it

Term	Full form	What it means	Why it matters to a data center
PUE	Power Usage Effectiveness	Total site energy ÷ IT equipment energy (1.0 is perfect)	Cooling and losses multiply every kWh. CAS savings apply to the whole site's draw

Carbon

Term	Full form	What it means	Why it matters to a data center
CO_2	Carbon dioxide	The main greenhouse gas from burning coal and gas	What climate targets and customers ask you to cut
CO_2eq	CO_2 equivalent	Other greenhouse gases converted to CO_2's warming effect	Most grid data, including Electricity Maps, is reported in CO_2eq
g, kg, t	Gram, kilogram (1,000 g), tonne (1,000 kg)	Mass of CO_2	Annual reports use tonnes; CAS logs kg per job
gCO_2/kWh	Grams of CO_2 per kilowatt-hour	Carbon intensity: how dirty each unit of grid electricity is, hour by hour	The number CAS minimises. India averages 727; it falls at midday and rises in the evening
tCO_2/MWh	Tonnes of CO_2 per megawatt-hour	The same measure in larger units (0.727 tCO_2/MWh = 727 gCO_2/kWh)	Used by CEA and in sustainability reports
Fossil share		Share of electricity from coal and gas in a given hour	The real-data validation measures this directly
Average vs marginal emissions		Average: the whole grid mix. Marginal: the plant that actually ramps when you add or remove load	Marginal is the truer measure of a saving; CAS uses average today (section 8)

Money and tariffs

Term	Full form	What it means	Why it matters to a data center
₹/kWh, paise	Rupees per kilowatt-hour; paise = 1/100 rupee	The price of each unit of electricity	60 paise/kWh off at midday is ₹19.20 on a 32 kWh job

Term	Full form	What it means	Why it matters to a data center
INR/MWh, USD/MWh	Rupees or US dollars per megawatt-hour	Power-exchange price units (₹2,500/MWh = ₹2.50/kWh)	The ML model forecasts these hourly prices
ToD / ToU	Time of Day / Time of Use tariff	Different prices for different hours	The reason clean hours are also cheap hours in India
HT, LT	High Tension, Low Tension	Supply voltage class; large sites are on HT	Data centers are usually HT consumers
HTP-I	High Tension Power, category I (Gujarat)	Gujarat's general HT tariff category	Used for CAS's ₹ calculations
LTMD	Low Tension Maximum Demand (Gujarat)	Tariff for LT consumers with load above 40 kW	Smaller server rooms and offices
FPPAS	Fuel and Power Purchase Adjustment Surcharge	A variable charge added for fuel costs	Not yet in CAS's ₹ figures (section 8)
IEX	Indian Energy Exchange	India's main power exchange; prices change every hour	Sites buying power on the exchange can save about 33% by timing (ML results)

Organisations and grid

Term	Full form	What it means	Why it matters to a data center
CEA	Central Electricity Authority	India's technical power regulator; publishes the grid emission factor	The official 727 g/kWh used for reporting
GERC	Gujarat Electricity Regulatory Commission	Sets Gujarat's tariffs	Source of the ToD rates CAS uses
DISCOM	Distribution Company (e.g. DGVCL, MGVCL, PGVCL, UGVCL, Torrent Power)	The utility that bills you	Its tariff decides the ₹ saving

Term	Full form	What it means	Why it matters to a data center
Grid-India (formerly POSOCO)	Grid Controller of India (formerly Power System Operation Corporation)	Runs India's national grid; publishes demand and generation data	Source of the measured validation data and price data
IN-WE	India, Western region (Electricity Maps zone code)	Gujarat, Maharashtra, Madhya Pradesh, Chhattisgarh, Goa	The zone CAS forecasts for Gujarat
BRSR	Business Responsibility and Sustainability Report	SEBI-mandated ESG disclosure for large listed Indian companies	CAS's audit log provides auditable emission-reduction evidence
IST, UTC	Indian Standard Time, Coordinated Universal Time	IST = UTC + 5:30	Grid data often comes in UTC; tariffs are in IST

Technology

Term	Full form	What it means	Why it matters to a data center
CAS	Carbon-Aware Scheduler	This system	—
GPU	Graphics Processing Unit	The chip that runs AI training and inference	The main power load in an AI data center
ML	Machine learning	Software that learns patterns from data	Forecasts carbon and price where no official forecast exists
API	Application Programming Interface	A way for software to fetch data automatically	How CAS gets live grid data
MAE	Mean Absolute Error	Average size of a forecast's miss, in the signal's own units	How the forecasts were compared
MAPE	Mean Absolute Percentage Error	MAE as a percentage of the true value	The same comparison as a percentage

Term	Full form	What it means	Why it matters to a data center
Checkpointing		Saving a job's progress so it can stop and resume	Lets CAS split long training runs across several clean windows

Sources

- CEA CO_2 Baseline Database: FY 2023-24 grid emission factor 0.727 tCO_2/MWh (SolarQuarter)
- Ember data on India's midday vs evening fossil generation (Down To Earth)
- Electricity (Rights of Consumers) Amendment Rules 2023: ToD tariff (Indian Infrastructure)
- GERC tariff schedule, DGVCL/MGVCL/PGVCL/UGVCL, from 1 Apr 2026
- GERC FY 2026-27 tariff order: solar-hours rebates (DeshGujarat)
- Wood Mackenzie: India data-center capacity to 12 GW by 2030 (Indian Infrastructure)
- Electricity Maps API documentation (free-tier endpoints)
- National Grid ESO Carbon Intensity API: live 48 h forecast used in testing
- Grid-India hourly demand, solar and wind, Sep 2021 – Jun 2025 (Mendeley Data, CC BY 4.0)
- HPE SustainCluster: carbon intensity, price and weather data used to train the ML model (GitHub, MIT)
- Electricity Maps pricing
- E2E Networks pricing (GPU and CPU, INR)